# LAN Design for a Corporate Office

**Scenario**

Designing a Scalable LAN for TechSavvy Inc.

Overview: You are a network engineer hired by TechSavvy Inc., a rapidly growing tech startup known for its innovative software solutions and tech-driven approach. The company is moving into a new office building to accommodate its expanding team and increasing operational demands. Your task is to design a robust and scalable Local Area Network (LAN) for this new office space. The network must support the current needs of the company and be flexible enough to handle future growth and technological advancements.

Company Background: TechSavvy Inc. specialises in developing cutting-edge software solutions for various industries, including finance, healthcare, and e-commerce. The company has grown rapidly over the past few years, increasing its workforce from 20 to 150 employees. As a tech-centric company, TechSavvy relies heavily on a reliable and efficient network infrastructure to support its operations, including software development, testing, customer support, and daily business communications.

**Task 1: Project Requirements Analysis.**

This section analyses the project requirements for TechSaavy Inc.

TechSaavy is a leading company that provides cutting edge software solutions for various industries.

The company has recently moved to an office building that spans three floors. The company will initially start with 150 employees with plans to expand that number to reach 300 employees.

Each department requires different needs to ensure efficiency and the flow of work.

The departments that the company covers involve ( development, HR and Finance, and Management).

Developers in the development department require high speed internet connection with low latency to ensure such tasks are completed smoothly like coding, testing tools, and file transferring. When the HR and Finance department needs strong security measures to be implemented for securing employees and company sensitive data. While the management requires strong network connection in meeting rooms for video conferencing with minimal lags to ensure stable video conferencing.

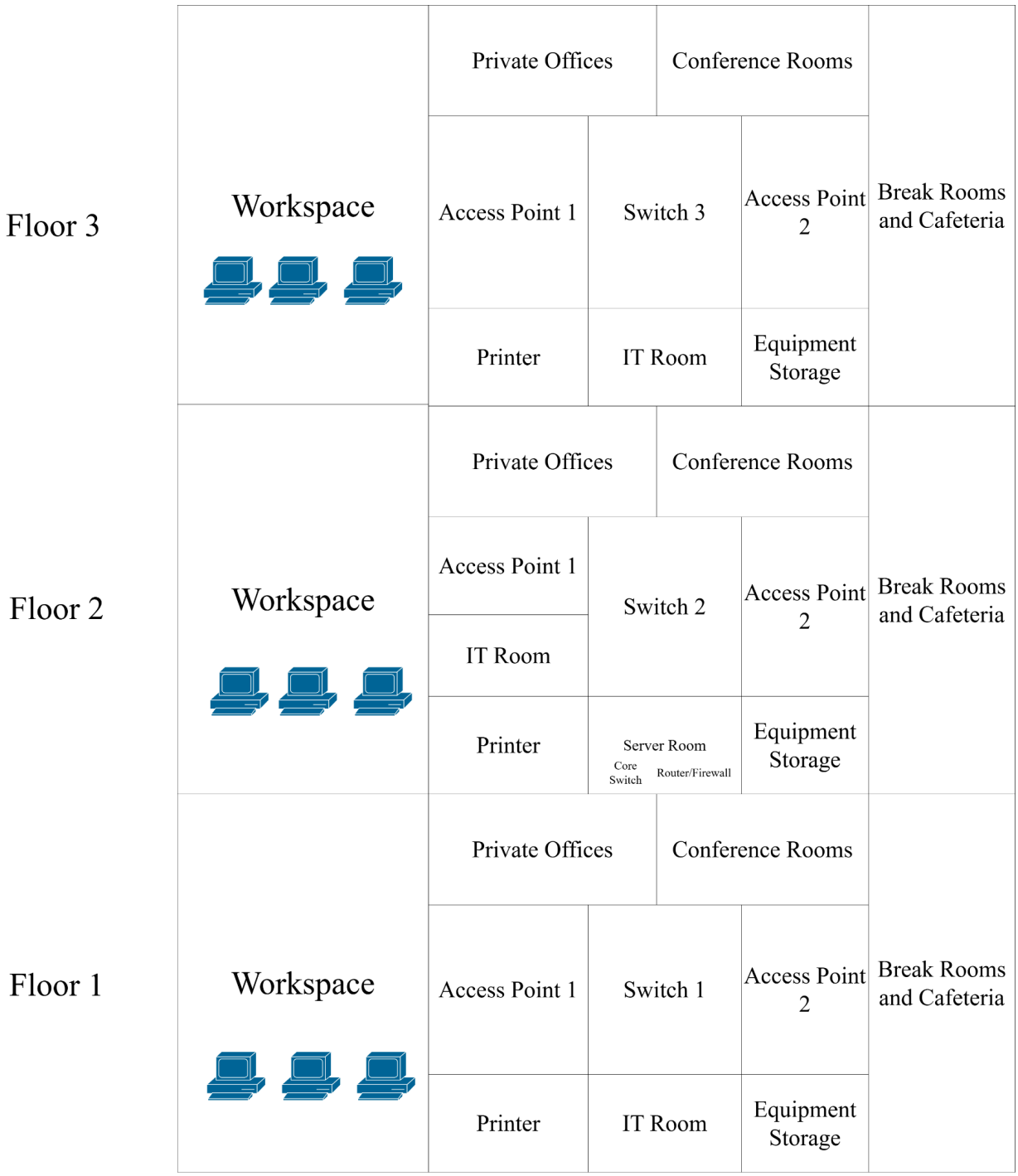
The building office will have various and wide range of devices that will be connected to the network. These devices involve PCs and laptops, mobiles and tablets, monitors for the IT staff members, Smart TVs in the conference room for video conferencing, routers and switches , and scanners and printers. Also, a guest wifi could be included to the network for optimal hospitality of visitors.

Since the company plans to embrace 300 employees, the design of the network should also be scalable for the future growth in order to get upgraded easily. However, network traffic could be expected in the long term when it comes to the expansion of employees. These network traffics involve transferring large amounts of data and files between the departments,video conferencing and

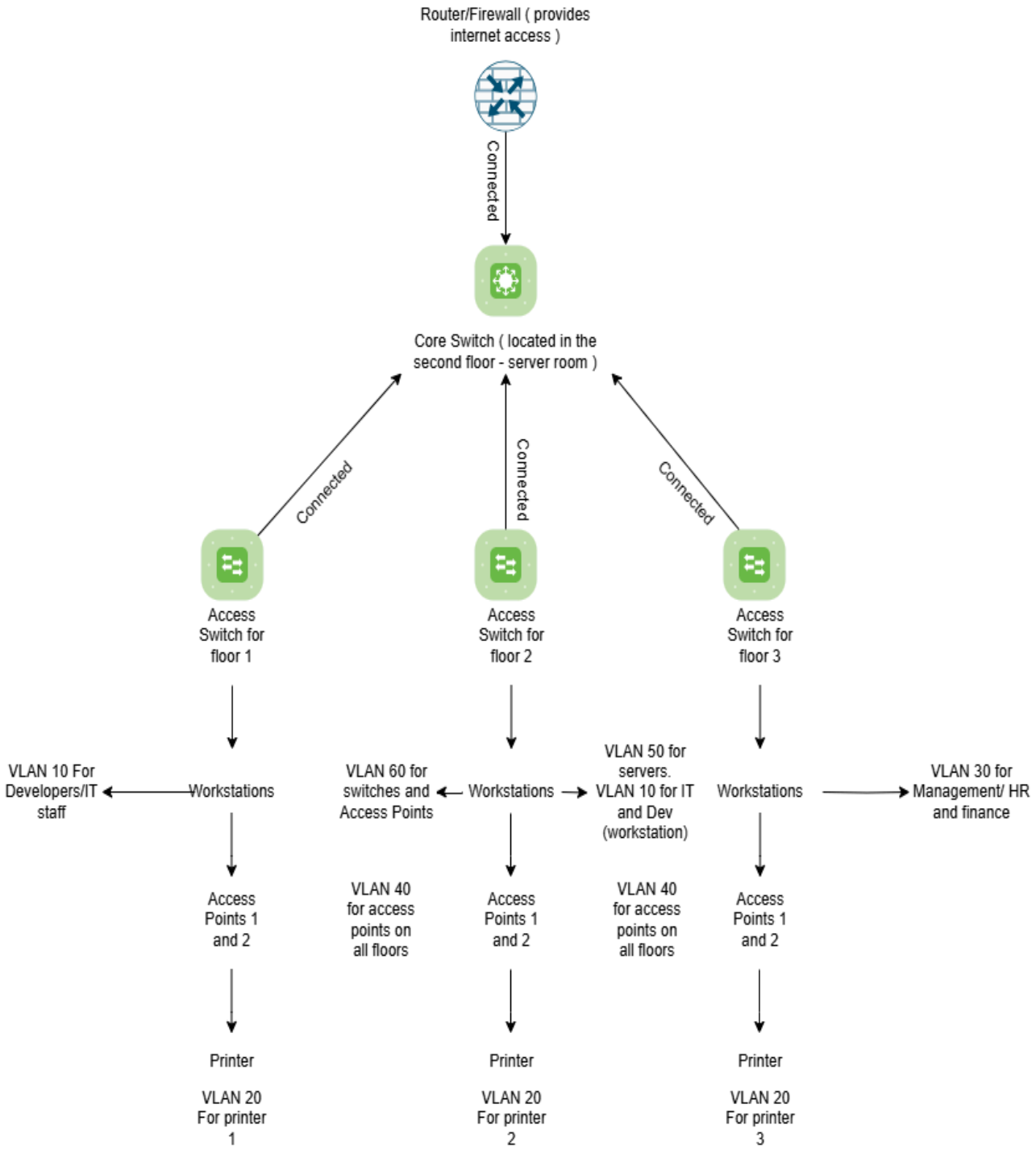
voice over ip because it requires to be stable to ensure better communication, and the large number of devices that could join the network such as laptops, mobiles, and tablets.

**Task 2: Network Topology Design.**

This section explores the appropriate network design for the new office building.



The provided diagram above illustrates a full floor plan for the new office building of the company, in this diagram, the layout shows how different departments within the company are distributed into a separate floor throughout the office building to minimize distraction between employees. Each floor contains separate workspace, private offices, conference rooms, break rooms and cafeteria, and storage rooms. This distribution of workplace could help in increasing the overall productivity of the employees. Additionally, the placement of switches in the three floors is considered strategic and significant because it helps in achieving better data flow and reduces latency over the LAN. The existence of server room on the second floor ensures full server coverage for the entire office building.This layout is also scalability supported, allowing for future growth in terms of implementing advanced technologies and expansion of employees. After designing the floor plan, a network topology will be made and shown to demonstrate how devices in different departments connect to ensure efficient data flow.


Router/Firewall ( provides internet access )
Connected
Core Switch ( located in the second floor - server room )
Connected
Connected
Connected
Access Switch for floor 1
Access Switch for floor 2
Access Switch for floor 3
VLAN 10 For Developers/IT staff
Workstations
VLAN 60 for switches and Access Points
Workstations
VLAN 50 for servers. VLAN 10 for IT and Dev (workstation)
Workstations
VLAN 30 for Management/ HR and finance
Access Points 1 and 2
VLAN 40 for access points on all floors
Access Points 1 and 2
VLAN 40 for access points on all floors
Access Points 1 and 2
Printer
VLAN 20 For printer 1
Printer
VLAN 20 For printer 2
Printer
VLAN 20 For printer 3

This diagram shows the network topology designed for the company. The network has a star architecture, with three edge switches one on each floor connected to a single core switch in the server room (Floor 2). Every device, including workstations, printers, and access points, is linked to the edge switch on its corresponding level.

To control internet access and maintain security, a router/firewall is connected to the main switch. The network segmentation was successfully configured using Virtual Local area network.this segmentation separates each department  to avoid network traffic. This design supports the company plans for future growth as it allows them to expand the number of employees and also manage the network traffic between departments.

Why choose specific VLANs IDs ?

Because each id separates each department to avoid traffic. For example, VLAN chosen for the IT and Developers department keeps them apart from sensitive departments such as the HR and finance. Additionally, the one assigned with management and HR/ Finance ensures that they are protected in terms of sensitive information and keeps them isolated from other departments. For printers, the VLAN chosen for them minimizes the risks and avoids user traffic. The VLANs assigned with the access points on all floors are dedicated to visitors as it is a guest wifi. This keeps the internet access for visitors isolated from the internal network of the company. Moreover, The chosen VLAN for the servers secures the communication between the nodes. Finally .The chosen VLAN for Switches and Access points is dedicated to IT administrators to manage the network and avoid network exposures.

**Task 3: Equipment Selection and Specification.**

This section explores the selection of equipment used to design the network and why they were chosen.

According to the floor plan, the distribution of departments was made across the office building. hence, a network design must support this distribution and guarantee company demands such as scalability and security

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Major equipment used were core switch, edge switches (access switches), routers, and access points.

Core switch ( which is located in the server room, particularly in the second floor) is responsible for connecting all floors together. It provides high speed connection. It can also handle high traffic between floors. Additionally, Edge switches are distributed in each floor (each floor has its own switch ). Edge switch is a floor level connector that connects users devices.

Router, which is connected to the core switch, is a device that provides internet access to users. It also provides secure connection because it includes a firewall built in. Moreover, access points were designed and distributed across the office building ( each floor has 2 access points ) to provide wifi coverage that spans the entire floor. In addition, a server device (that is located in the server room) is a device that provides internal services for the company. It allows employees to share files and access data and applications.

**Task 4: IP Addressing and Subnetting.**

This section covers the development of an IPv4 addressing scheme for the network.

| VLAN | Departments | Subnets | Subnet Masks | Gateway | Usable IPs |
|---|---|---|---|---|---|
| 10 | Dev/IT | 10.10.10.0/24 | 255.255.255.0 | 10.10.10.1 | 254 |
| 20 | Printers | 10.10.20.0/25 | 255.255.255.128 | 10.10.20.1 | 126 |
| 30 | HR/Fin/Mngt | 10.10.30.0/25 | 255.255.255.128 | 10.10.30.1 | 126 |
| 40 | Guest WiFi | 10.10.40.0/24 | 255.255.255.0 | 10.10.40.1 | 254 |
| 50 | Server | 10.10.50.0/25 | 255.255.255.128 | 10.10.50.1 | 126 |
| 60 | Infrastructure | 10.10.60.0/26 | 255.255.255.192 | 10.10.60.1 | 62 |

The provided table shows different departments and their assigned ip addresses. Each department has its own VLAN, subnets, Subnet masks, and gateways. In order to manage network traffic efficiently, each VLAN was assigned to a particular subnet by using private IP range ( class A ) 10.0.0.0/16. VLAN 10, which was designed for developers and the IT department, was given the subnet 10.10.10.0/24 to support approximately 100 users with the space for future growth. Printers were given the subnet 10.10.20.0/25 because it provides enough ip addresses for the existing printers and the future printers that could be implemented.

The departments of management, finance, and human resources were assigned 10.10.30.0/25 . This guarantees that private information is protected within a secure and segmented subnet. guest wifi section uses 10.10.40.0/24 for guests to access Wi-Fi. This ensures that guest traffic remains isolated from internal systems and supports up to 254 users.

In order to provide a safe and manageable address space for hosting business services, VLAN 50 was isolated and dedicated for servers and given the subnet 10.10.50.0/25. Finally, VLAN 60, which was designed to manage switches and access points, uses 10.10.60.0/26, which is more than enough for the network's infrastructure devices and helps in limiting administrator access to only authorized IT staff. To make routing and device configuration easier, each subnet uses the first usable IP address (such as.1) as the default gateway.

**Task 5: Network Security Design.**

This section covers the designing of the security infrastructure for the network.

After designing the network, it is significant to ensure the internal data of the company is well secured and easy to access by employees. Hence, security measures were implemented to the network design to ensure sensitive data is secured. especially in sensitive departments such as HR and finance.

In the network design, each department was assigned its own VLAN to isolate traffic and prevent unauthorized access. For example, the guest Wi-Fi network works on a separate VLAN to ensure that visitors cannot access internal systems. additionally, Wireless communication is secured using WPA2 encryption and MAC address filtering could be applied to prevent unauthorized device access.

The router has a built in firewall to protect the network from external threats. It filters both entering and exiting traffic and blocks unauthorized connections using access control rules. In addition, Access Control Lists are implemented between VLANs to prevent direct access to sensitive resources.

Finally, The switch supports port security, which helps prevent unauthorized physical access by limiting the number of devices per port and disabling unused ports. It also connects all the VLANS and each VLAN is connected to its wireless access point. This structure allows the employees to move freely around the office.

**Task 6: Network Management and Monitoring.**

This section outlines the management and monitoring strategies for the network.

Designing a network requires constant monitoring in terms of performance and security to ensure that services are always available for the employees, company's operations run smoothly, and the data is always secured and retrieved any time. To ensure that the network is reliable, tools and softwares must be implemented to achieve reliability.

1- Monitoring and testing the network:

> According to linkedin.com, Network monitoring and testing are essential tools for improving [Company's] network reliability. They help [The Company] to collect and analyze data about .. network performance, health, and behavior. [they] can use network monitoring and testing to detect and troubleshoot problems, such as congestion, errors, outages, and security breaches. [they] can also use them to measure and optimize.. network efficiency, capacity, and quality. Network monitoring and testing can be done using various methods, such as SNMP, ping, traceroute, or netstat .

2- Network Redundancy:

Network redundancy and backup are strategies for increasing [the] network reliability by adding extra resources and paths to [the] network. Network redundancy means having multiple connections, devices, or servers that can take over if one fails. Network backup means having a copy of [the] network data and configuration that can be restored if needed. By adopting network redundancy and backup, [they] can reduce the impact of network failures and ensure business continuity.

3- Network Security Practices:

Network security is a critical aspect of network reliability, as it protects [the] network from external and internal threats that can compromise [the] network integrity, availability, and confidentiality. Network security best practices include encrypting [the Company's] network traffic, using strong passwords and authentication methods, updating [the] network software and firmware, implementing firewalls and antivirus software, and educating [the] network users and administrators about cyber risks and policies.

4- Updating and reviewing the network constantly:

Network reliability is not a one-time task, but a continuous process that requires regular review and update. [the] network needs and goals may change over time, as well as the network environment and technology. Therefore, [the company] should review and update [the] network regularly to ensure that it meets [the company's] current and future demands. [they] should also document [the] network changes and keep track of [the] network performance and issues.

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